

*Same as well
Just better*

SAME YOU SINCE 2009
JUST BETTER

Before we start

Let's talk about your
reading assignment

(The Strategy Question): In your own words, what is the main difference between GitFlow and Trunk-Based Development?

(The Conflict Question): According to the readings, what are the specific visual markers (symbols) that Git injects into a file to show you exactly where a merge conflict happened?

(The PR Question): Why do engineering teams use Pull Requests instead of just letting developers merge their own code directly into the main branch?

(The SDLC Question): According to the Atlassian SDLC guide, what are the first two phases that must happen before any code is actually written?

(The Golden Rule Question): Based on the branching strategies and the video, why is it considered dangerous (and often restricted) to commit your daily unfinished work directly to the main branch?

Lesson 4

Collaborative Development (Branching, Conflicts & PRs)



- **git branch <name>**: Creates a new branch.
- **git checkout <name> (or git switch)**: Moves you to that branch.
- **git checkout -b <name>**: Creates AND moves to the branch in one command.

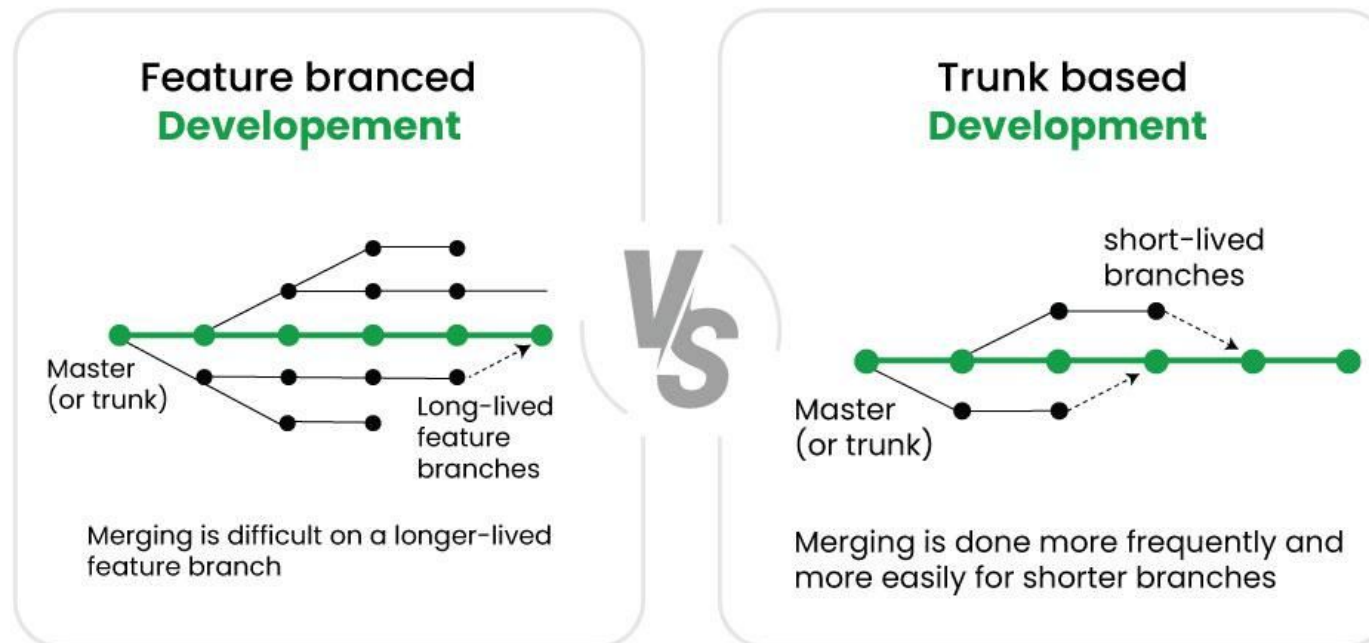
```
# How to Create a Branch in GIT?

# Syntax
$ git checkout -b <new-branch>

# Example
$ git checkout -b feature-2
Switched to a new branch 'feature-2'
```

Trunk-Based: Everyone merges to main frequently (usually daily). Requires strong automated testing.

GitFlow: Strict structure. main (production), develop (integration), plus feature, release, and hotfix branches.





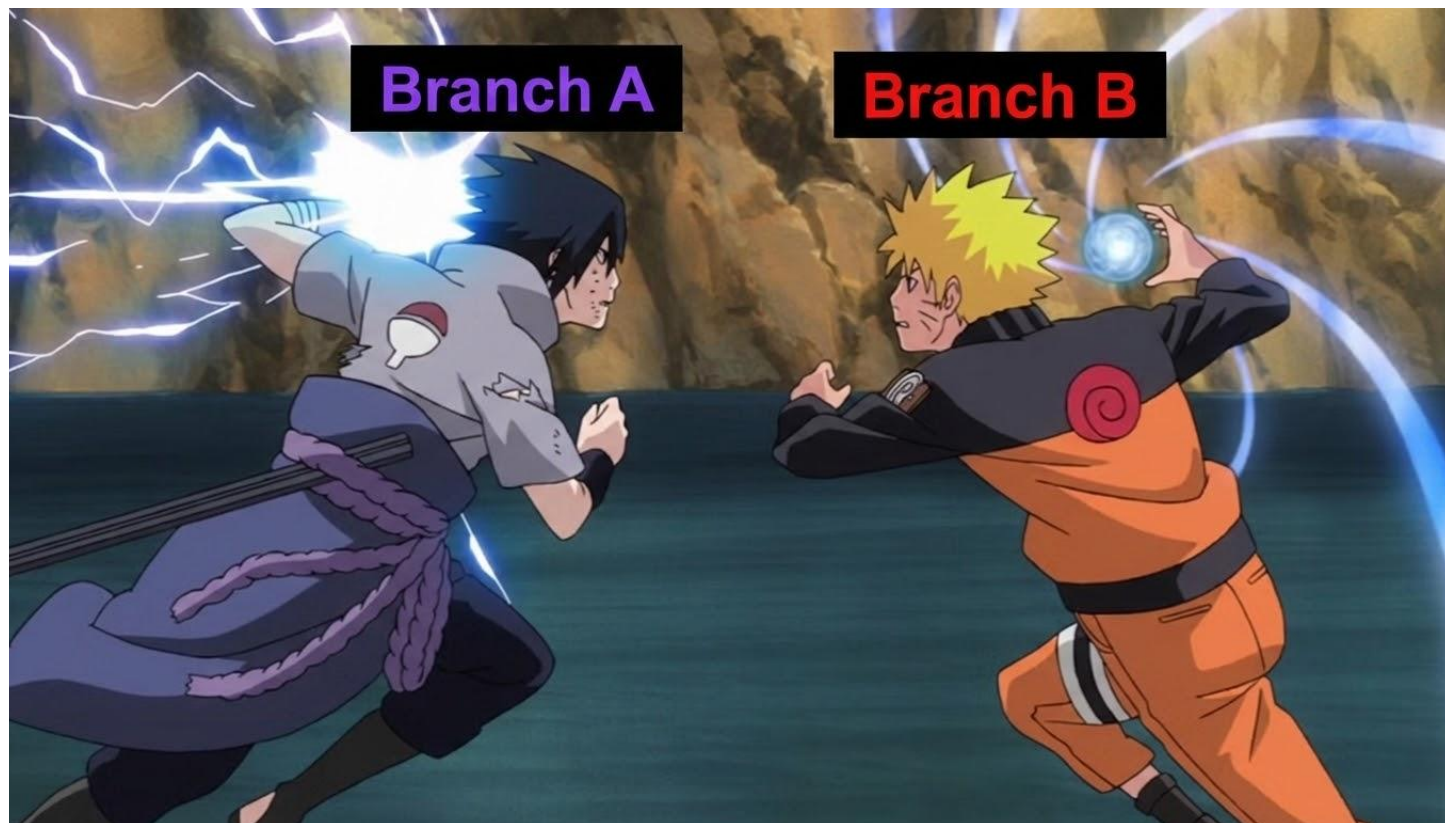
A PR (or Merge Request in GitLab) is a request to merge your branch into the main codebase.

It's the foundation of Code Review.

Allows team members to comment, approve, or request changes before code goes live.



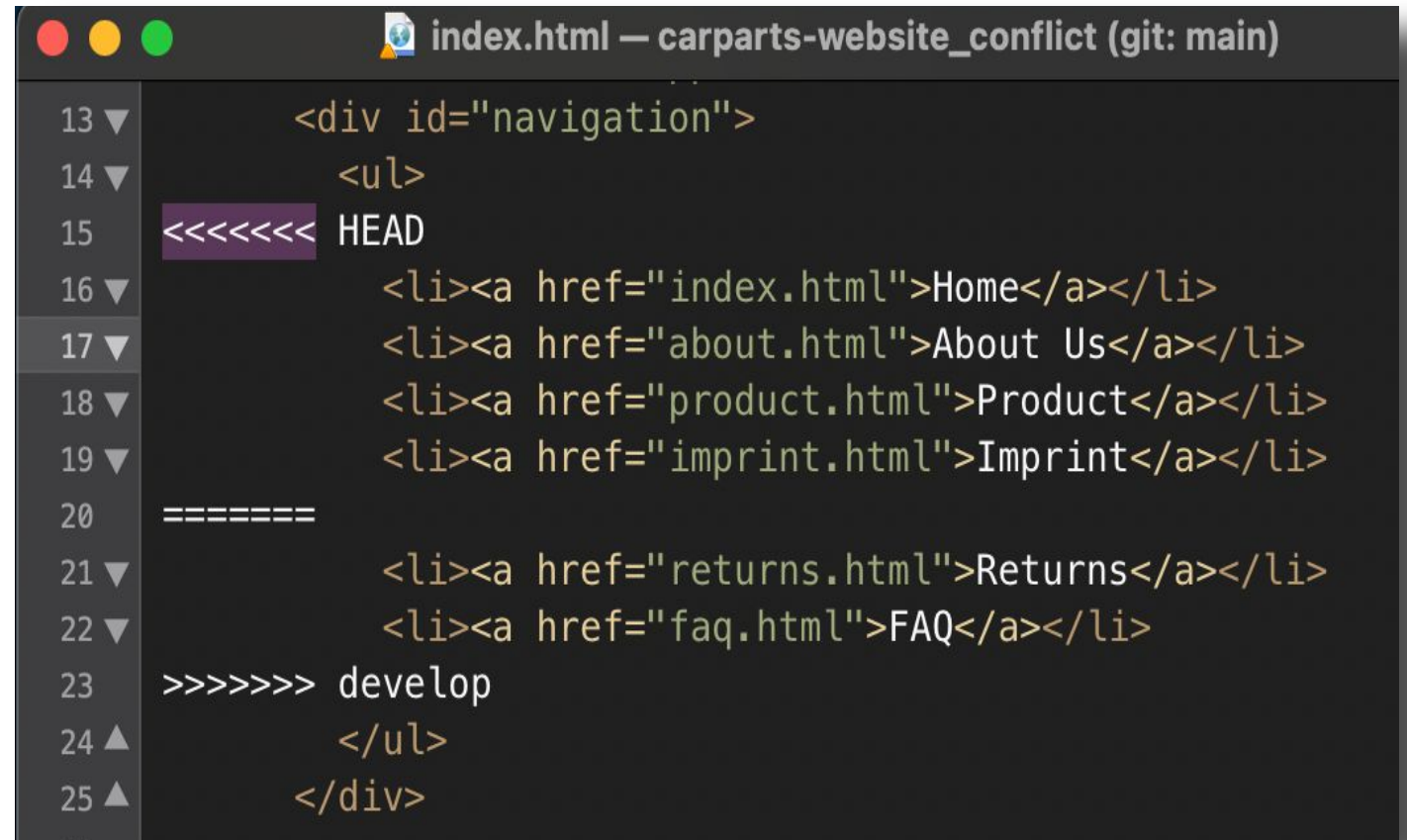
- Conflicts happen when two branches change the exact same line in the exact same file.
- Git pauses the merge and says: "I don't know which version to keep. You choose."



<<<<<< HEAD: The current state of the branch you are merging into.

=====: The divider.

>>>>>> feature-branch: The changes coming from your feature branch.



```
13 <div id="navigation">
14   <ul>
15   <<<<<< HEAD
16     <li><a href="index.html">Home</a></li>
17     <li><a href="about.html">About Us</a></li>
18     <li><a href="product.html">Product</a></li>
19     <li><a href="imprint.html">Imprint</a></li>
20   =====
21     <li><a href="returns.html">Returns</a></li>
22     <li><a href="faq.html">FAQ</a></li>
23   >>>>>> develop
24   </ul>
25 </div>
```

The Fix: Delete the markers, keep the code you want, save, git add, and git commit.



- **Reviewers:** Attack the code, not the person. Suggest alternatives ("Have you considered awk here?").
- **Authors:** Don't take it personally. Provide context in your PR description.
- **LGTM = Looks Good To Me!**



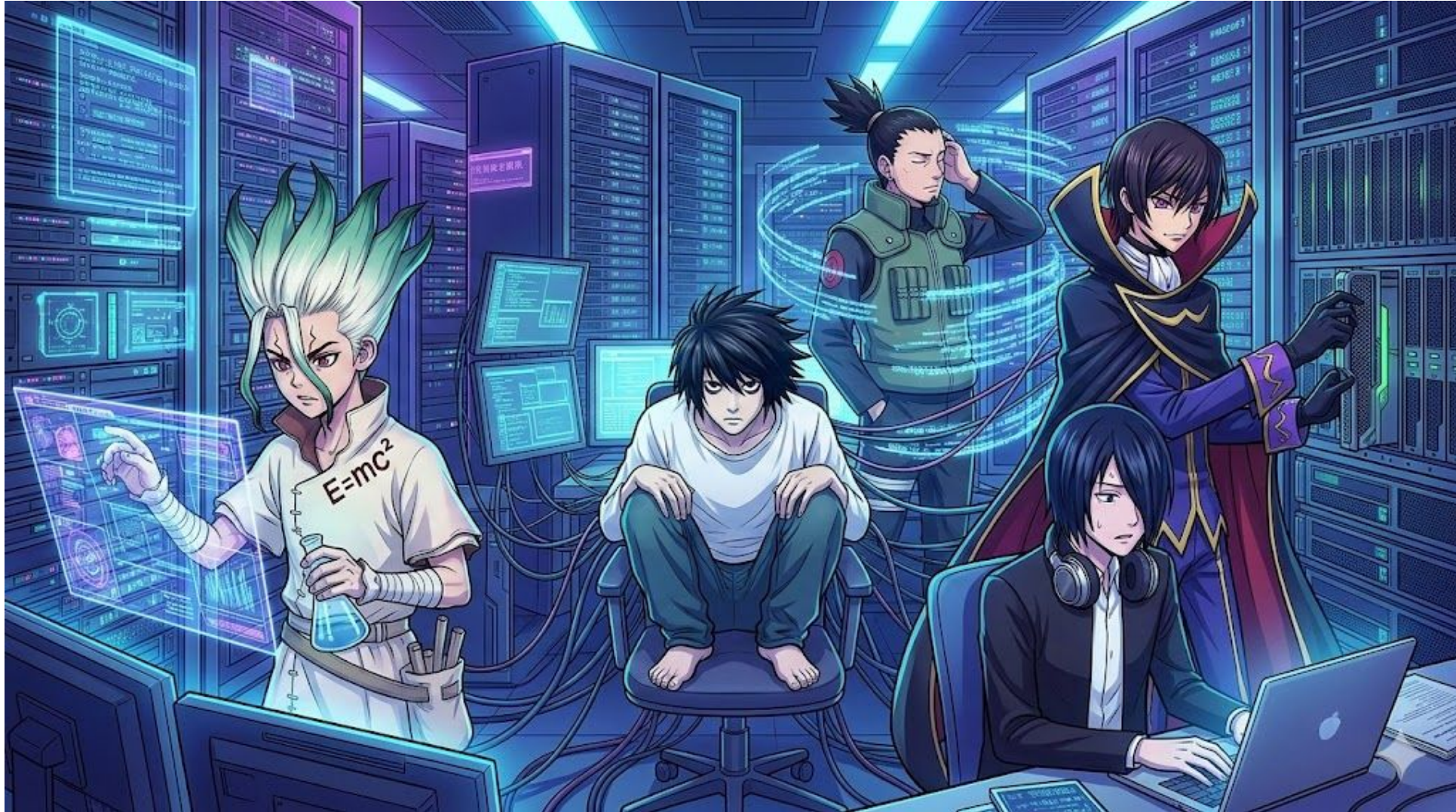
What does "LGTM" mean?

LGTM is an acronym, frequently used when code reviews on GitHub.

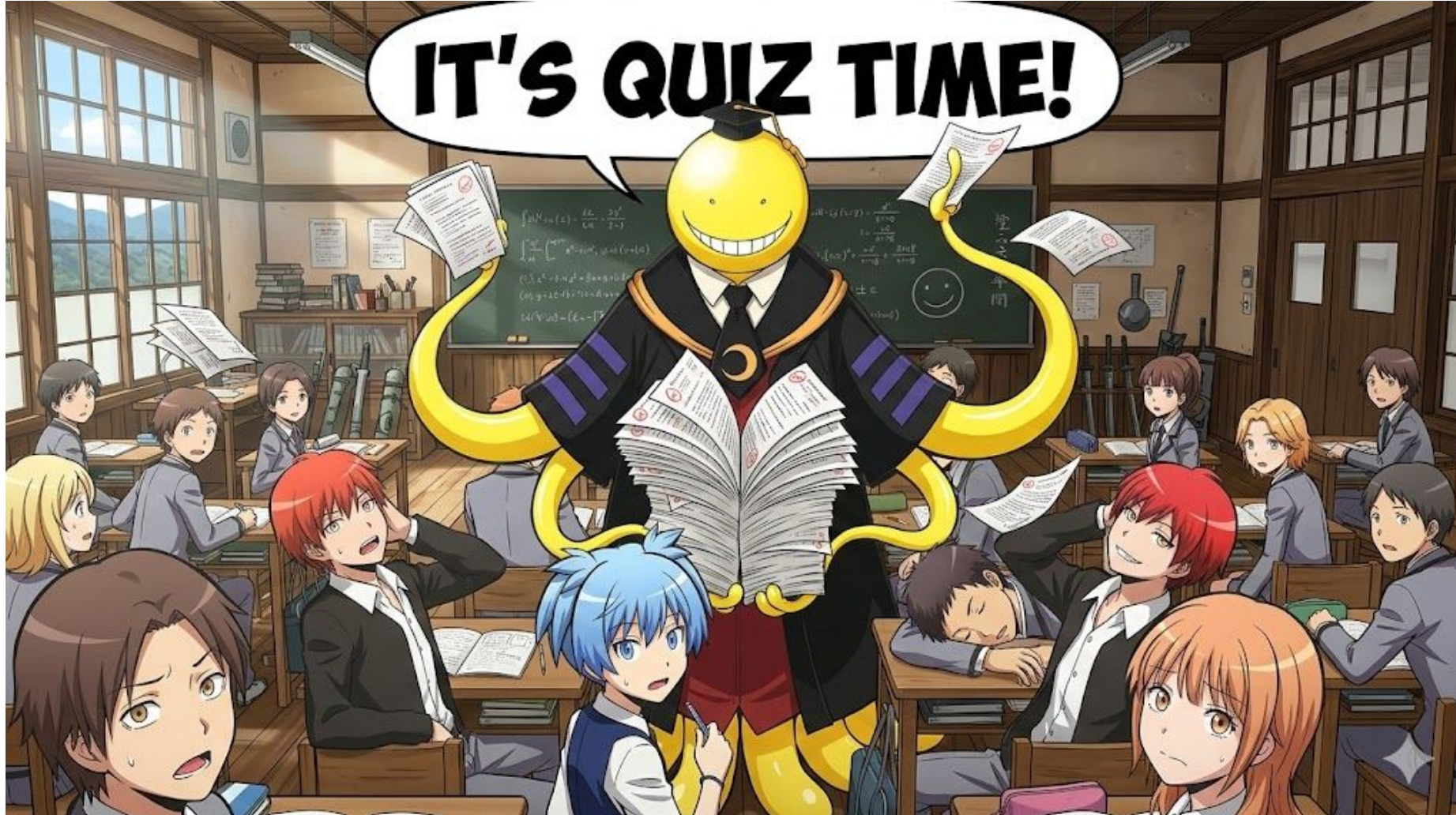
The interpretation were:

- Looks Good To Me. (Because Google says so.)
- Let's Get This Merged.
- Legitimate
- Let's Gamble, Try Merging.

Now, let's break things together.



Now, it's time for the Quiz



- Homework 4: Team Simulation (Solo).
- Next Lesson: Python for DevOps Part 1.
- Prep Work: Ensure Python 3 is installed on your local machines.

Questions?

CONTACT US